

Literature-implied partial answer

The operator-norm case of the convexification Mazur map question

Status. `literature_implied_answer` (partial subcase).

Source question. In Section 4 of Javier Alejandro Chavez-Dominguez, *Uniform homeomorphisms between spheres of unitarily invariant ideals*, arXiv:2305.07169, Theorem `thm-homeomorphism-convexification` proves that for $3 \leq p < \infty$, and every 1-symmetric sequence space E , the map

$$G_p(x) = u|x|^p, \quad x = u|x|,$$

is a uniform homeomorphism from $\mathcal{S}(S_{E^{(p)}})$ onto $\mathcal{S}(S_E)$. The following remark says that it is not known whether the restriction $p \geq 3$ is actually needed, and that the restriction is inherited from Jocić's Theorem 3.1.

Supporting theorem. In Section 10 of A. B. Aleksandrov and V. V. Peller, *Operator Holder-Zygmund functions*, arXiv:0907.3049, Theorem `fcc` proves that if $0 < \alpha < 1$ and $f \in \Lambda_\alpha(\mathbb{R})$, then for self-adjoint operators A, B and a bounded operator R ,

$$\|f(A)R - Rf(B)\| \leq C_\alpha \|f\|_{\Lambda_\alpha} \|AR - RB\|^\alpha \|R\|^{1-\alpha}.$$

This is an operator-norm theorem for quasicommutators.

Identification. Fix $p > 1$ and set $\alpha = 1/p$. The function

$$f(t) = \text{sign}(t)|t|^{1/p}$$

belongs to $\Lambda_{1/p}(\mathbb{R})$. Applying the Aleksandrov–Peller quasicommutator estimate gives

$$\|f(A)R - Rf(B)\| \leq C_p \|AR - RB\|^{1/p} \|R\|^{1-1/p}.$$

Replacing B by $-B$ turns the denominator into the anticommutator: for $A, B \geq 0$,

$$\|A^{1/p}R + RB^{1/p}\| \leq C_p \|AR + RB\|^{1/p} \|R\|^{1-1/p}.$$

After substituting $A = x^p$ and $B = y^p$, this is precisely the operator-norm version of the missing reverse estimate in Lemma `lemma-sum` of arXiv:2305.07169.

The source paper already proves the forward uniform continuity of G_p for all $p \geq 1$. The only obstruction to extending Theorem `thm-homeomorphism-convexification` below $p = 3$ is the inverse uniform-continuity estimate for the self-adjoint/anticommutator reduction. The estimate above supplies it for the operator norm.

Conclusion. For the 1-symmetric sequence space $E = c_0$, so that $S_E = \mathcal{K}(\ell_2)$ with the operator norm and $E^{(p)} = c_0$, the map G_p is a uniform homeomorphism

$$G_p : \mathcal{S}(S_{c_0}) \longrightarrow \mathcal{S}(S_{c_0})$$

for every $1 < p < \infty$. Thus the restriction $p \geq 3$ is not needed in this operator-norm subcase.

Limitations. This packet does not solve the full question for arbitrary unitarily invariant ideals and $1 < p < 3$. The supporting theorem used here is in operator norm; it does not provide the corresponding estimate for an arbitrary symmetric ideal norm S_E .

References.

- J. A. Chavez-Dominguez, *Uniform homeomorphisms between spheres of unitarily invariant ideals*, arXiv:2305.07169.
- A. B. Aleksandrov and V. V. Peller, *Operator Holder-Zygmund functions*, arXiv:0907.3049.